

Project Methodology Summary

Group 14

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1.Introduction and Research Question

The London Fire Brigade responds to over 100,000 incidents a year, yet fewer than one in ten is an actual fire. This project repositions LFB incident data as an urban sensor, a longitudinal record capable of tracing how structural forces such as demographic ageing, climate stress, and public sector restructuring have reshaped emergency demand between 2014 and 2025. Drawing on urban resilience, we argue that each incident type maps to a distinct pressure on the city: fires to building safety, flooding to climate adaptation, forced entries to care gaps in an ageing population, and agency assists to systemic shifts in public service delivery. The research is organized around three escalating questions.

(1) How has the composition of LFB incidents shifted, and has the Brigade effectively become a multi-function emergency service?

(2) Is growing Special Service demand concentrating in outer London, where station coverage is thinnest?

(3) Do different incident types follow distinct seasonal rhythms that overlap in ways a flat-staffed system cannot absorb?

2.Datasets

2.1 Primary Dataset: LFB Incident Records (2009 to 2026)

1,952,560 incident records from three LFB open data releases on the London Datastore: a CSV for 2009 to 2017 (988,279 records) and two Excel files covering 2018 to 2023 (670,635) and 2024 onwards (293,646). Each record holds 36 fields, including incident category (IncidentGroup, SpecialServiceType, StopCodeDescription), temporal markers (CalYear, DateOfCall, HourOfCall), spatial identifiers (ProperCase borough, IncGeo_WardCode, Easting_rounded and Northing_rounded at 50m precision), PropertyCategory, IncidentStationGround, and FirstPumpArriving_AttendanceTime.

2.2 Supplementary Datasets

(1) London ward boundary shapefile, 2018 vintage (LondonDataStore, Statistical GIS Boundary Files for London), used for the seasonal ward map. The 2018 vintage matches the ONS ward codes used by LFB from 2018 onwards.

(2) Fire station locations (Open Data Institute, theodi/FNR_Analysis), used to map the 102 currently active stations after removing the 10 stations closed in 2014 and one duplicate water station; per-station incident counts come from the LFB IncidentStationGround field.

(3) GLA Ward Profiles and Atlas (LondonDataStore), used as a secondary reference for borough demographic context.

3.Data Cleaning & Processing

All processing was conducted in Python with pandas, geopandas, pyproj and shapely. The three source files were concatenated after harmonising column names and data types. Records missing CalYear, date, or Easting and Northing were dropped from spatial analysis; borough labels were standardised to the official GLA convention (for example “And” to “and”, “Upon” to “upon”).

Different chapters use different temporal windows. The yearly composition, three trajectories, 250m grid density layers and seasonal ward map use 2014 to 2025 (post-station-closure window). The response time grid uses 2009 to 2025 to maximise sample size per cell. Fire station incident totals use 2018 to 2025, the period over which IncidentStationGround is consistently populated. Partial-year 2026 records are excluded throughout.

For all grid layers, each incident's Easting_rounded and Northing_rounded were floored to a 250m British National Grid cell, then reprojected from EPSG 27700 to WGS84 with pyproj. Cells with fewer than three incidents were dropped, and colour scales clip at the 95th percentile to prevent outlier cells from flattening the rest of the map. The seasonal ward layer was built by spatially joining incident points to the 2018 ward polygons in EPSG 27700, then aggregating by ward code and season.

4.Methods and Tools

4.1 Technology Stack

(1) Python (pandas, geopandas, shapely, pyproj, scipy): data cleaning, aggregation, coordinate transformation, 250m grid generation, spatial joins, and Pearson correlation testing

(2) JavaScript (vanilla ES6+) and D3.js v7: interactive visualisation; D3 handles the animated Special Service slope chart (stroke-dasharray transitions) and the borough scatter plot

(3) Chart.js v4: stacked area, horizontal bar, and doughnut charts

(4) Mapbox GL JS: 250m grid density maps (95th-percentile colour capping), synchronised triple-map view, response time heatmap, and bivariate map (fire density × response time, 3×3 colour matrix)

(5) Scrollama and GSAP/ScrollTrigger: scrollytelling triggers and chapter entry animations

(6) Three.js r128: 3D particle sphere backgrounds between chapters

(7) GitHub Pages: single-page HTML deployment

4.2 Spatial Analysis

The site combines spatial analysis at three resolutions.

(1) Four 250m*250m grid layers cover the Greater London BNG extent (easting 500,000 to 565,000; northing 155,000 to 205,000) and aggregate fire density, false alarms, special service density, and average first pump attendance time, all reprojected to WGS84 via pyproj. Cells with fewer than three records are dropped and colour scales clip at the 95th percentile.

(2) The Chapter 2 bivariate map encodes fire density and response time on the same 250m cells, classifying each variable into tertiles client-side and mapping the combinations onto a 3×3 colour matrix. The response time grid additionally overlays the 102 active fire station points.

(3) Borough-level analysis drives the special service growth versus response time scatter (with a danger zone annotation for outer boroughs facing both pressures) and the inner versus outer London divergence chart, using the GLA inner London statistical definition.

(4) Ward-level seasonal analysis spatially joins incident points to the 2018 LondonDataStore ward shapefile and aggregates by ward, season, and incident type.

4.3 Visualization Design

The project is delivered as an interactive scrolling website, structured around three chapters:

Ch1: The Identity Shift: From Fire-Fighting to Multi-Function Emergency Service

Ch2: The Spatial Mismatch: Demand Decentralises, Infrastructure Stays Put

Ch3: The Temporal Comparison: Seasonal Peaks the System Cannot Absorb

The site is a three chapter scrollytelling narrative on a dark background with orange and teal accents. Chapter transitions are interleaved with Three.js 3D particle scenes that visually bridge each chapter shift. The reader moves from incident composition (Chapter 1) to spatial mismatch (Chapter 2) to seasonal stacking (Chapter 3), with each chapter combining a sticky graphic and scroll-triggered text. Interaction is structural rather than decorative: a year slider with play button cycles the triple density maps from 2014 to 2025, sub-category pills filter each grid by incident type, a draggable side panel ranks the slowest and fastest boroughs with name search, a station toggle overlays the 102 active stations on the response time map, and a custom five-button type fan on the Chapter 3 ward map switches between incident types under summer and winter views. Animations are timed to data transitions, not page decoration.

5.Limitation

(1) The analysis is descriptive, not causal: outer London demand growth coincides with the 2014 closures, but population change, climate, ageing, and inter-agency demand remain uncontrolled confounders.

(2) LFB coordinates are rounded to 50m and some early records carry only postcode-level precision, so cell assignments hold uncertainty where postcodes straddle boundaries.

(3) The 250m grid is one resolution without sensitivity testing; finer or coarser cells

could change which patterns surface.

(4) Response time is averaged across all incident types per cell, so a cell mean may reflect the local mix rather than geographic accessibility.

(5) The seasonal map defines summer as May to September and winter as November to March, leaving April and October unclassified and masking shoulder dynamics.

(6) The inner versus outer split uses a custom 14 borough definition, one of several defensible standards, so the divergence magnitude is partly definitional.